

Tuza's conjecture for line and total graphs

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Abstract

For a graph G , let $\nu_\Delta(G)$ be the maximum number of pairwise edge-disjoint triangles and let $\tau_\Delta(G)$ be the minimum number of edges meeting every triangle. We prove $\tau_\Delta(G) \leq 2\nu_\Delta(G)$ when G is the line graph or the total graph of any finite simple graph. The proofs use edge-disjoint incidence cliques and a two-coloring of root edges obtained from Euler tours. A local packing adjustment handles the exceptional Euler-tour seam for line graphs; adjoining each original vertex to the minority color of its incidence clique absorbs the seam for total graphs. The factor two is sharp over line graphs. A cover constructed from distinct incident-edge representatives gives, for every cubic triangle-free graph F , the exact identity $\tau_\Delta(\text{Tot}(F)) = \nu_\Delta(\text{Tot}(F)) = 5|V(F)|/2$. The proof constructions presented here follow the accompanying Lean formalization, including its explicit finite clique-packing certificates.

1 Introduction

A *triangle packing* in a graph G is a collection of pairwise edge-disjoint triangles, and a *triangle transversal* is a set of edges meeting every triangle. Their maximum and minimum cardinalities are denoted by $\nu_\Delta(G)$ and $\tau_\Delta(G)$, respectively. Tuza conjectured that

$$\tau_\Delta(G) \leq 2\nu_\Delta(G) \tag{1}$$

for every finite graph G [5].

We establish (1) for two graph operators built from the incidence structure of a root graph. The *line graph* $L(H)$ has vertex set $E(H)$, with two vertices adjacent when the corresponding root edges meet. The *total graph* $\text{Tot}(F)$ has vertex set $V(F) \cup E(F)$; two elements are adjacent when they are adjacent or incident in F .

Our main results are the following.

Theorem 1.1. *If H is a finite simple graph, then*

$$\tau_\Delta(L(H)) \leq 2\nu_\Delta(L(H)).$$

The factor two cannot be improved over the class of line graphs.

Theorem 1.2. *If F is a finite simple graph, then*

$$\tau_\Delta(\text{Tot}(F)) \leq 2\nu_\Delta(\text{Tot}(F)).$$

Corollary 1.3. *If F is a finite simple cubic triangle-free graph, then*

$$\tau_\Delta(\text{Tot}(F)) = \nu_\Delta(\text{Tot}(F)) = \frac{5}{2}|V(F)|.$$

The two proofs share an edge decomposition. Incidences at a root vertex induce a complete graph, and these complete graphs have pairwise disjoint edge sets. Elementary complete-graph packings give local lower bounds. A single coloring of the root edges controls triangles that cross several incidence cliques. In total graphs we also select all original root edges; the corresponding bridge triangles supply a compatible packing contribution. A Hall assignment gives a sharp cover when the root is cubic and triangle-free.

The line-graph inequality for triangle-free roots appears in [2, Corollary 3.3.3]; Theorem 1.1 removes that hypothesis. Nearby results establish Tuza's inequality under a maximum average degree bound [4]; a recent preprint treats maximum degree at most seven [1]. Other results improve the coefficient for certain K_4 -free graphs [3]. Those hypotheses do not cover the unbounded-degree classes considered here. We are not aware of a previous statement of either theorem in the full scope given above.

All graphs in the paper are finite and simple; empty and disconnected roots are allowed. Simplicity ensures that two distinct root edges have at most one common endpoint. The proofs construct an actual transversal C and packing $\mathcal{P}$ in the same graph with $|C| \leq 2|\mathcal{P}|$. The complete-graph bound uses a three-part construction and strong induction, with explicit certificates for orders below 21. These finite certificates are checked by Lean's kernel; no exact formula for complete-graph packing numbers is assumed. The results concern the two stated graph classes and do not resolve Tuza's conjecture for arbitrary graphs.

2 Complete-graph tools

For $d \geq 0$, put

$$t(d) = \binom{d}{2} - \left\lfloor \frac{d^2}{4} \right\rfloor, \quad D(d) = \nu_{\Delta}(K_d).$$

A balanced bipartition of $V(K_d)$, followed by the selection of every edge inside a part, gives a triangle transversal of size $t(d)$. Explicitly,

$$t(2r) = r(r-1), \quad t(2r+1) = r^2. \quad (2)$$

We use the following elementary lower bounds on $D(d)$. The basic complete-graph inequality is classical; compare the construction attributed to Sebő in [2, Theorem 3.3.2]. The following proof uses the same Latin-square recurrence and finite bases as the Lean development and supplies the additional even-order slack.

Lemma 2.1 (Local clique slack). *For every integer $d \geq 0$,*

$$t(d) \leq 2D(d). \quad (3)$$

If $d \geq 6$ is even, then

$$t(d) + 2 \leq 2D(d). \quad (4)$$

Proof. For $q > 0$, partition $3q$ vertices into three copies A, B, C of $\mathbb{Z}/q\mathbb{Z}$. Take the q^2 triangles

$$\{a_A, b_B, (a+b)_C\}, \quad a, b \in \mathbb{Z}/q\mathbb{Z}.$$

Any two coordinates determine the third, so distinct triangles share at most one vertex. They use only edges between parts. Adding a maximum packing inside each of A, B, C therefore gives

$$D(3q) \geq q^2 + 3D(q). \quad (5)$$

Also, $D(a) \leq D(b)$ for $a \leq b$, by embedding K_a into K_b .

For completeness, the following describes precisely the finite witnesses used for $0 \leq n < 21$, on vertices $0, \dots, n-1$. Use the empty list for $n \leq 2$. For $n \geq 3$, put $r = \lfloor n/2 \rfloor$ and begin with

$$\{\{i, j, r + ((i + j) \bmod r)\} : 0 \leq i < j < r\}.$$

If $n = 2r \geq 6$, add $\{r, r+1, r+2\}$. If $n = 2r+1$, add

$$\{2r, i, r + ((2i) \bmod r)\} \quad (i \in I_r), \quad I_r = \begin{cases} \{0, \dots, r-1\}, & r \text{ odd}, \\ \{0, \dots, r/2-1\}, & r \text{ even}. \end{cases}$$

Each listed set has three distinct vertices. For the initial triangles, sharing a vertex in each of the two halves forces the remaining index to agree. The extra even-order triangle uses only edges in the second half. For odd orders, multiplication by two is injective on I_r modulo r ; moreover, an initial triangle containing i cannot also contain $r + ((2i) \bmod r)$, since its other first-half index would then equal i . Thus every pair of distinct listed triangles shares at most one vertex. Their sizes p_n are

n	0	1	2	3	4	5	6	7	8	9	10
p_n	0	0	0	1	1	2	4	6	7	8	11

and

n	11	12	13	14	15	16	17	18	19	20
p_n	15	16	18	22	28	29	32	37	45	46

Direct substitution verifies $t(n) \leq 2p_n$ and, for even $n \geq 6$, $t(n) + 2 \leq 2p_n$. The formal proof checks the same literal lists for three-element cardinality, pairwise intersection size at most one, and these numerical bounds.

Proceed by strong induction. For $n \geq 21$, write $n = 3q + s$ with $q \geq 7$ and $0 \leq s < 3$. The function t is nondecreasing, and

$$t(3q + s) + 2 \leq 2q^2 + 3t(q). \quad (6)$$

Indeed it suffices to take $s = 2$. When $q = 2r$, the right side minus the left side is $2r^2 - 6r - 2 \geq 0$ for $r \geq 4$; when $q = 2r+1$, it is $2r^2 - 4r - 4 \geq 0$ for $r \geq 3$. The induction hypothesis, (5), and monotonicity now give

$$t(n) + 2 \leq 2q^2 + 3t(q) \leq 2(q^2 + 3D(q)) \leq 2D(3q) \leq 2D(n).$$

This proves both assertions, including the required even-order slack. $\square$

3 Line graphs

Let H be a finite simple graph. For $v \in V(H)$, let W_v be the set of edges of $L(H)$ corresponding to unordered pairs of distinct root edges incident with v . Two adjacent edges of a simple graph have a unique common endpoint, and hence

$$E(L(H)) = \dot{\bigcup}_{v \in V(H)} W_v. \quad (7)$$

We call W_v the *wedge packet* at v .

Lemma 3.1 (Line-graph triangle classification). *Every triangle of $L(H)$ is exactly one of the following:*

- (i) *a star triangle formed by three root edges incident with one vertex v , whose three edges all belong to W_v ;*
- (ii) *a root triangle formed by the three edges of a triangle xyz in H , with one edge in each of W_x, W_y, W_z .*

Proof. Three distinct pairwise intersecting two-element sets either have a common element or are $\{x, y\}, \{x, z\}, \{y, z\}$. Applied to three pairwise adjacent root edges, these are precisely the two stated cases. $\square$

The star triangles in W_v form the triangle system of $K_{d_H(v)}$. Choose $D(d_H(v))$ edge-disjoint star triangles at every root vertex. The packet partition (7) shows that their union is a packing of size

$$P_0 = \sum_{v \in V(H)} D(d_H(v)). \quad (8)$$

Write

$$T = \sum_{v \in V(H)} t(d_H(v)). \quad (9)$$

Lemma 2.1 gives $T \leq 2P_0$.

The cover uses a balanced red-blue coloring of the root edges. We isolate the required form.

Lemma 3.2 (Alternating Euler coloring). *Let K be a connected graph.*

- (a) *If K has an odd-degree vertex, its edges can be colored red and blue so that the two color degrees are equal at every even-degree vertex and differ by one at every odd-degree vertex.*
- (b) *If every degree of K is even and $|E(K)|$ is even, the two color degrees can be made equal at every vertex.*
- (c) *If every degree of K is even and $|E(K)|$ is odd, the two color degrees can be made equal except at one chosen seam vertex, where they differ by two.*

Proof. For (a), add a new dummy vertex adjacent to every odd-degree vertex. The augmented graph is connected and Eulerian. Start an Euler tour at the dummy vertex and color successive tour edges alternately. At every original vertex the tour edges pair into differently colored entering and leaving edges. Any odd cyclic seam occurs at the dummy vertex. After deleting the dummy edges, the stated balances remain.

For (b) and (c), an isolated vertex needs no coloring. Otherwise alternately color an Euler tour of K . If the tour length is even, every visit pairs two different colors. If it is odd, the first and last edges have the same color and the unique discrepancy of two occurs at the starting vertex. $\square$

Given such a coloring, select in W_v every wedge whose two root edges have the same color. Every star triangle is hit by the pigeonhole principle. Every root triangle is hit because two of its three root edges have the same color. If the red and blue degrees at v are a and b , the local cost is

$$\binom{a}{2} + \binom{b}{2}.$$

By (2), this equals $t(d_H(v))$ when an even degree is balanced or an odd degree differs by one. At a difference-two even degree it equals $t(d_H(v)) + 1$.

Consequently, every connected component has a cover of cost at most its contribution to T , except possibly a connected Eulerian component with an odd number of edges. Such a component has a cover with one extra edge. The following lemma absorbs this unique surcharge.

Lemma 3.3 (Odd Eulerian seam repair). *Let K be a connected Eulerian component of H with an odd number of edges. Put*

$$T_K = \sum_{v \in V(K)} t(d_H(v)), \quad P_K = \sum_{v \in V(K)} D(d_H(v)).$$

There is a triangle transversal C_K of $L(K)$ and a triangle packing $\mathcal{P}_K$ in $L(K)$ such that $|C_K| \leq 2|\mathcal{P}_K|$.

Proof. The alternating Euler cover has size at most $T_K + 1$. We distinguish three exhaustive cases.

If some vertex has degree at least six, its degree is even and (4) supplies two units of local packing slack. Together with (3) at every other vertex, this gives $T_K + 1 \leq 2P_K$.

Suppose next that K is triangle-free. Discard the global coloring. At each vertex, split its incident root edges into two equal parts and select the wedges lying inside a part. Since there are no root triangles, these local selections cover every triangle and have total size exactly T_K . Thus $T_K \leq 2P_K$.

It remains that every positive degree is two or four and that K contains a root triangle xyz . Keep the alternating Euler cover of size at most $T_K + 1$. At a degree-two vertex the local packing is empty. At a degree-four vertex, $D(4) = 1$, and a triangle of the local K_4 can avoid any one prescribed wedge: if that wedge is ab , take the triangle on $\{a, c, d\}$. We may therefore choose the local maximum packings at x, y, z to avoid, independently, the three wedges of the triangle in $L(K)$ corresponding to xyz . Adding it to the local packing gives a packing of size $P_K + 1$. Hence

$$T_K + 1 \leq 2P_K + 1 \leq 2(P_K + 1).$$

These cases are exhaustive because, in the absence of a degree at least six, Eulerianity forces every positive degree to be two or four, after which the component is either triangle-free or contains a root triangle. $\square$

Proof of Theorem 1.1. Apply the alternating Euler cover to every component not requiring the seam repair, and Lemma 3.3 to every connected Eulerian odd-edge component. Line graphs and their triangle systems are disjoint across root components, so the component covers and packings may be united. This proves $\tau_\Delta(L(H)) \leq 2\nu_\Delta(L(H))$.

For sharpness, $L(K_{1,4}) \cong K_4$ has $(\nu_\Delta, \tau_\Delta) = (1, 2)$. Indeed, every two of its triangles share an edge, one edge cannot meet all four triangles, and two opposite edges do. $\square$

Remark 3.4. The quantifier in Lemma 3.3 is one surcharge per connected Eulerian odd-edge component, not per block, root triangle, or high-degree vertex. Odd cycles show why the triangle-free branch must discard the Euler seam rather than charge it to a nonexistent local packing.

4 Total graphs

Let F be a finite simple graph. Write

$$d_v = d_F(v), \quad m = |E(F)|.$$

For every $v \in V(F)$, put

$$Q_v = \{v\} \cup \delta_F(v),$$

where $\delta_F(v)$ is the set of root edges incident with v , and an edge in this set is regarded as its edge-vertex in the total graph. The set Q_v induces K_{d_v+1} . Its edges are the incidence edges from v to its incident edge-vertices and the wedge edges between pairs of root edges meeting at v . These packet edge sets are pairwise disjoint. Original root edges lie outside all of them. For a root edge uv , let $B_{uv} = \{u, uv, v\}$ be its *bridge triangle*; here uv inside the vertex set denotes the edge-vertex.

Proof of Theorem 1.2. Apply Lemma 3.2 componentwise to color the root edges red and blue. At every odd-degree vertex the color degrees differ by one, and at every even-degree vertex they differ by at most two.

In each Q_v , retain these colors on the edge-vertices and color the original vertex v with the less frequent color, breaking ties arbitrarily. Adding one vertex to the minority side changes a discrepancy of zero, one, or two into one, zero, or one, respectively. Thus Q_v is balanced and its within-color edges number exactly $t(d_v + 1)$.

Let C consist of these within-color edges in every Q_v , together with all m original root edges. This gives a single red-blue coloring of all vertices of $\text{Tot}(F)$: each edge-vertex keeps its root-edge color, and each original vertex has its chosen minority color. Every monochromatic edge of $\text{Tot}(F)$ belongs to C . Indeed, an original-original edge was selected outright, an incidence edge belongs to its original vertex's packet, and a wedge belongs to the packet at the unique common root endpoint. Every triangle has a monochromatic edge by the pigeonhole principle, so C is a triangle transversal. Counting the packet contributions and original edges gives

$$|C| \leq m + \sum_v t(d_v + 1). \quad (10)$$

For a packing, take all m bridge triangles and, at each v , a packing of $D(d_v)$ triangles among the edge-vertices in $\delta_F(v)$. Distinct bridges use distinct original and incidence edges. The additional triangles use only wedge edges, whose packets are mutually disjoint. Thus their union is one packing $\mathcal{P}$ of size

$$|\mathcal{P}| = m + \sum_v D(d_v) \leq \nu_\Delta(\text{Tot}(F)). \quad (11)$$

The identity and degree-sum bound

$$t(d+1) - t(d) = \left\lfloor \frac{d}{2} \right\rfloor, \quad \sum_v \left\lfloor \frac{d_v}{2} \right\rfloor \leq m \quad (12)$$

follow from (2) and the degree sum. Therefore

$$\tau_\Delta(\text{Tot}(F)) \leq |C| \leq 2m + \sum_v t(d_v) \leq 2|\mathcal{P}| \leq 2\nu_\Delta(\text{Tot}(F)),$$

where the penultimate inequality is Lemma 2.1. □

5 Cubic triangle-free roots

For a cubic triangle-free root, a distinct choice of one incident edge at each vertex gives a smaller cover. We first prove the assignment used in the formal development.

Lemma 5.1 (Distinct incident-edge representatives). *If a finite simple graph F has minimum degree at least two, there is an injective map $p : V(F) \rightarrow E(F)$ such that $p(v)$ is incident with v for every vertex v .*

Proof. Apply Hall's theorem to the vertex-edge incidence bipartite graph. For $S \subseteq V(F)$, let $N(S)$ consist of the root edges incident with at least one vertex of S . Counting incidences gives

$$2|S| \leq \sum_{v \in S} d_F(v) \leq 2|N(S)|,$$

because a root edge has at most two endpoints in S . Hence $|S| \leq |N(S)|$ for every S , which is precisely Hall's condition. $\square$

Proof of Corollary 1.3. Put $n = |V(F)|$. Then $m = 3n/2$ and $D(3) = 1$, so the packing in (11) has $3n/2 + n = 5n/2$ triangles.

Choose p by Lemma 5.1. At each v , write its three incident edges as $p(v), a_v, b_v$. In the packet $Q_v \cong K_4$, select the incidence edge $vp(v)$ and the wedge $a_v b_v$. These two edges form a perfect matching of Q_v , so every triangle in the packet meets one of them. Also select every original edge outside $p(V(F))$.

For a bridge B_e , if $e = p(v)$ for some v , its selected incidence edge $vp(v)$ meets the bridge; otherwise its original edge is selected. Since F is triangle-free, every total triangle is a packet triangle or a bridge. To see this directly, three original vertices would give a root triangle; two original vertices force their joining edge-vertex; one original vertex forces the other two edge-vertices into its packet; and three edge-vertices are classified by Lemma 3.1, whose root-triangle case is excluded. This is the same case analysis used in the formal cover proof. The selection is therefore a transversal. Its size is at most

$$2n + (m - |p(V(F))|) = 2n + (m - n) = m + n = 5n/2,$$

using the injectivity of p . Finally, every transversal contains a distinct edge for each member of an edge-disjoint packing, so

$$\frac{5n}{2} \leq \nu_\Delta(\text{Tot}(F)) \leq \tau_\Delta(\text{Tot}(F)) \leq \frac{5n}{2}. \quad \square$$

6 Formal verification

The accompanying Lean repository implements the constructions presented above. The local clique proof uses the Latin packing, the same finite lists for orders below 21, and the same strong-induction inequality. The line and total proofs use the Euler colorings, packet unions and seam repair described here; the cubic cover uses the incident-edge assignment of Lemma 5.1. The earlier balanced-orientation refinement is retained in a separate informal note in the repository and is not a result of this revised paper.

The four headline declarations are

Theorem 1.1	<code>Tuza.lineGraph_satisfiesTuza</code>
Line-graph sharpness	<code>Tuza.lineGraph_factor_sharp</code>
Theorem 1.2	<code>Tuza.totalGraph_satisfiesTuza</code>
Corollary 1.3	<code>Tuza.cubic_triangleFree_total_parameters</code>

The formal graph definitions use ordinary finite simple graphs, edge-disjoint triangle packings and edge transversals. Empty and disconnected roots are included. The cubic conclusion uses the equivalent natural-number identities $2\tau_{\Delta} = 5|V(F)|$ and $2\nu_{\Delta} = 5|V(F)|$.

At source commit `82f6bb5`, using Lean 4.33.1 and the pinned Mathlib dependency, all four declarations passed Palomar’s Comparator check and replay by Lean and NanoDa. Only propositional extensionality, classical choice and quotient soundness occur in the audited axioms. The deliberately incomplete Challenge statements are separate from the completed Solution proofs. A detailed proof-to-declaration map and full commit identifiers are provided in the repository. That mechanical run checked the Lean sources before this expository revision; it did not review this revised manuscript, establish novelty, or constitute human expert review or completed registration.

7 Concluding remarks

The two operator classes permit globally compatible certificates because adjacent root edges have a unique common endpoint. In line graphs, the only possible loss in an Euler coloring is one extra selected wedge per connected Eulerian component with an odd number of edges. In total graphs, the original vertex balances its enlarged incidence clique, while bridge triangles pay for the selection of original root edges.

The factor two is sharp for line graphs. We do not determine the best factor for total graphs, although the cubic triangle-free case has equality between packing and transversal numbers. Extensions to multigraph roots, generalized line graphs, or quasi-line graphs require additional arguments and are not claimed here.

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